Alex Nguyen

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EDUCATION

The University of Texas at Austin, Austin, Texas — May 2025
B.S., Aerospace Engineering, Honors, GPA: 3.94
Relevant Courses:
Computational Fluid Dynamics, Rocket Science, Flight Dynamics, Structural Dynamics, Propulsion, Spacecraft Systems Engineering

PROFESSIONAL EXPERIENCE

Flare-X Fire Suppression Internship – Austin, TX — May – August 2025

- Led design and execution for important mechanisms in fire suppression payload drone, including the payload bay, hatched door mechanism, and lifting surface positioning
- Manufactured prototypes and visual demonstrations for designed prototypes using 3D printing to present to project managers and chief engineers
- Crafted renderings of vehicle and requirements for use by systems integration leads to allow for consistency between teams and systems

ATA Engineering Student Co-op – San Diego, CA — August – December 2024

- Worked on projects from varying fields like engineering design, entertainment, and rotor dynamics
- Drafted and validated over 35 different parts, weldments, and assemblies for renovation and documentation of attraction seen by thousands daily
- Designed and implemented code for state-space mechanics and airfoil dynamics for solver modules
- Modeled, edited, and fabricated 3D model of structures in completed space lander project for display
- Trained and studied static and dynamic modeling techniques for varieties of structures with NX and Nastran software

Aeroelasticity Research Under NASA ULI Program - Austin, TX — October 2022 – Present
Undergraduate Research Assistant – Under Dr. Jayant Sirohi

- Worked on FAST Technology for Hypersonic Vehicles project
- Designed Mach 5 wind tunnel model of hypersonic vehicle with preliminary testing technology, for strain and temperature
- Specialized in fabrication of wind tunnel model through FDM, SLA, and SLS methods
- Installed, calibrated, and set standard practice for Fiber Bragg Grating System on wind tunnel model
- Utilized FEM to validate static and dynamic load cases for said wind tunnel model
- Ran tests to measure strain on sections of tabletop sized model of experimental hypersonic vehicle
- Utilized modal analysis to predict deformation and strain from actuated forces through Ansys Workbench
- Formulated and coded Strain-to-Deformation algorithm using mode shape data analysis
- See “Research and Publications”

UT Austin Aerospace Department, Austin, TX — September 2022 – Present
Departmental Tutor

- Taught and assisted fellow aerospace and computational students in statics, dynamics, and engineering computation problems
- Reinforced academic strategies and assisted in the continued success of struggling peers

RESEARCH AND PUBLICATIONS

- Panigrahi, A., Nguyen, A. Q., Eitner, M. A., & Sirohi, J. (2024). *Strain-Data-Driven Force Reconstruction Using Pseudo-Inverse Matrix*. Conference Proceedings of the Society for Experimental Mechanics Series, 117–133. https://doi.org/10.1007/978-3-031-68188-2_12

EXPERIENCE

Texas Rocket Engineering Lab – Austin, Texas — September 2022 – May 2024
Diversity and Inclusion Officer, Launch Operations Team

- Formulated and maintained an inclusive and friendly laboratory environment
- Designed and conducted analysis to confirm strength requirements for lifting mechanism used in lifting rocket to upright position, as well as in launch rail considerations

Texas School for the Deaf Volunteer – Austin, Texas — August 2023 – Present

- Volunteered and contributed to events and efforts like the Homecoming Fair and its Book Fair
- Worked to build bonds in Deaf communities and learn about Deaf culture

SKILLS

Computer: MATLAB, SolidWorks, Python, Ansys Workbench, Simulink, NX/Simcenter3D, NX NASTRAN, CATIA, OpenFoam, Paraview
Languages: Fluent Vietnamese, Intermediate ASL

HONORS AND AWARDS

Houston Rodeo Scholarship Recipient ·Texas Interscholastic League Foundation Scholarship Recipient · National Merit Finalist · Linford Family Endowed Scholarship